

Aurora 3000 Calibration Guide

Standard Operating Procedure for Automated Full Calibration (Zero & CO₂ Span)

Hardware Note: Unlike filter-based photometers, the Aurora 3000 manages its own pneumatics entirely. It utilizes an internal zero-air pump and filter for the zeroing phase and an internal solenoid valve to route calibration gas for the span phase. **You do not need to disconnect the ambient sample line or attach a zero filter to the inlet.**

Phase 1: Preparation & Gas Delivery

Step 1.1: Prepare the Calibration Gas (CO₂)

Before initiating the sequence from the dashboard, you must ensure the span gas is ready to flow.

- Verify that the CO₂ span gas cylinder is properly connected to the **Span Inlet** on the top of the Aurora 3000.
- Open the main cylinder valve and adjust the delivery regulator.
- Ensure the gas flow rate is regulated to approximately **3 to 4 L/min** to adequately flush the measurement cell without causing over-pressurization.

Phase 2: Automated Calibration Sequence

Step 2.1: Initiate Full Calibration Routine

Trigger the chained sequence via the DAQ driver to perform both calibration steps automatically.

- Navigate to the instrument dashboard in your UI.
- Change the **Calibration Routine** setting to `full_cal`.
- *System Response:* The DAQ driver will command the instrument to jump to State 2 (Zero Calibration). The instrument will automatically halt the main sample pump and engage the internal zero-air pump.

Step 2.2: Monitor Sequencer Status

The DAQ driver will track the instrument's progress through the two phases of calibration.

- **Phase 1 (Zeroing):** The **Calibration Status** field will display **zeroing**. The instrument will run particle-free air until its internal stability criteria are met, adjust the zero offset, and return to normal mode.
- **Phase 2 (Spanning):** Once the zero phase completes, the driver automatically commands the instrument to jump to State 1 (Span Calibration). The **Calibration Status** field updates to **spanning**. The instrument opens the internal span valve to pull the CO₂ gas, waits for optical stability, adjusts the calibration curve, and returns to normal mode.

Note: Each phase typically requires 15 to 20 minutes to complete depending on the instrument's programmed stability threshold and timeout constraints.

Phase 3: Completion & Post-Calibration

Step 3.1: Verify Calibration Results

The sequence concludes autonomously when the instrument returns to State 0 (Normal Monitoring).

- **Success:** The **Calibration Status** will update to **success** and the **Calibration Routine** will revert to **none**.
- *System Response:* Because the Aurora manages its own states, it will automatically turn off the span solenoid, re-engage the main sample pump, and immediately resume ambient aerosol monitoring.

Step 3.2: Secure Gas Delivery System

Post-calibration cleanup is minimal but critical for safety and resource conservation.

- **Close the main valve** on the CO₂ span gas cylinder securely.
- No further interaction with the UI is required. The system is already logging ambient data using the freshly calculated calibration curve.

Important: Never leave the CO₂ or FM-200 gas cylinder valves open when the calibration is complete. While the internal solenoid is closed, leaving high-pressure lines continuously charged can cause slow leaks or condensation issues over time.